



FIGURE 12. Two $\#$ -moves convert $T(9, 2)$ (left) to the unknot (right).

the standard projection. It follows that $C(4_1) = 2$. Clark also showed that $C(K) \leq 2 \cdot g(K) + 1$ for any knot K and asked if this upper bound was ever achieved.

Clark's question was finally answered in 1995 by Murakami and Yasuhara [30] using the Gordon-Litherland pairing. To do this, they first proved the following necessary condition for a knot K to bound a $\beta_1 = 2$ nonorientable spanning surface F .

Theorem 4.1 (Murakami-Yasuhara [30]). *There is a pair of curves generating $H_1(F; \mathbb{Z})$ such that the Goeritz matrix of F takes the form:*

$$G_F(K) = \begin{bmatrix} l & m \\ m & n \end{bmatrix}$$

and satisfies $\sigma(K) = \text{sign}(G_F(K)) - (l + 2m + n)$, where l, n are odd and m is even.

Consider, as in [30], the knot 7_4 . This knot has 3-genus 1 and hence $C(7_4) \leq 3$. Since 7_4 is hyperbolic, we have $C(7_4) \geq 2$. It therefore suffices to show that $C(K) \neq 2$. Now, suppose that $C(K) = 2$. Since 7_4 has determinant 15, one can write out integral binary quadratic forms with determinant 15 (see e.g. Conway-Sloane [8]). Using a case analysis, it is then possible to show that no such binary quadratic form satisfies Theorem 4.1. Hence, it follows that $C(7_4) = 3 = 2 \cdot g(7_4) + 1$.

Crosscap numbers are in general difficult to compute. Yet progress is currently being made. For example, Kalfagianni-Lee [19] determined upper and lower bounds on the crosscap numbers for several families of knots in terms of the coefficients of the Jones polynomial.

4.3. Non-orientable slice genus. The (*smooth*) 4-genus of a knot K is the smallest genus among all surfaces properly embedded in $\mathbb{B}^4$ such that $\partial F = K$. A well-known lower bound on the 4-genus is $|\sigma(K)|/2$. In analogy with crosscap number, the (*smooth*) non-orientable slice genus $\gamma^*(K)$ of a knot K in $\mathbb{S}^3$ is defined as the smallest first Betti number of all compact connected non-orientable surfaces F smoothly embedded in $\mathbb{B}^4$ such that $\partial F = K$. If K is slice, then $\gamma^*(K)$ is defined to be 0. In 1975, Viro [39] showed that $\gamma^*(4_1) = 2$, and hence 4_1 does not bound a Möbius band in $\mathbb{B}^4$.

Another proof that $\gamma^*(4_1) = 2$ follows from a result of Yasuhara, which is proved using the Gordon-Litherland pairing. Recall that the quadratic form $q : H_1(F; \mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z}$ of a knot K is defined by $q(x) = x^T A x$, where F is a Seifert surface of K and A is a Seifert matrix for F . The Arf invariant of K is the Arf invariant of this quadratic form. Hence, $\text{Arf}(K) = 0$ if q maps a majority of elements of $H_1(F; \mathbb{Z}/2\mathbb{Z})$ to 0 and otherwise $\text{Arf}(K) = 1$. The Arf invariant is also the (mod 2) coefficient of z^2 in the Conway polynomial of K . Note that there is a well-defined injection $\mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/8\mathbb{Z}$ given by $x \rightarrow 4 \cdot x$.

Theorem 4.2 (Yasuhara [40]). *If a knot K in $\mathbb{S}^3$ bounds a Möbius band in $\mathbb{B}^4$, then*

$$\sigma(K) + 4 \cdot \text{Arf}(K) \equiv 0 \text{ or } \pm 2 \pmod{8}.$$